

Disproof of Conjecture 00000003824: the Nakajima monomial crystal $M(\lambda)$ is not $\bigoplus_{\mu \leq \lambda} B(\mu)$

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Abstract

Conjecture 00000003824 asserts that the Nakajima monomial crystal $M(\lambda)$ decomposes into connected components $\bigoplus B(\mu)$, where μ runs exactly over the dominant integral weights $\mu \leq \lambda$ (dominance order), each appearing exactly once. This is false already for $\mathfrak{sl}_2$ and $\lambda = 2$: the monomial crystal generated by the highest-weight monomial Y_0^2 is $\{Y_0^2, Y_0Y_1^{-1}, Y_1^{-2}\} \cong B(2)$, a single string with 3 vertices, while $B(2) \oplus B(0)$ has 4 vertices and a highest-weight vertex of weight 0. More generally $M(\lambda) \cong B(\lambda)$ for every λ , so the claim fails for every $\lambda \geq 2$. We also treat two alternative readings of “ $M(\lambda)$ ” (the union of the components of all highest-weight monomials of weight λ , and the crystal of products of λ fundamental monomials); $\lambda = 3$ defeats all three readings simultaneously. The $\mathfrak{sl}_2$ counterexample, the failure for all $\lambda \geq 2$, and the three readings at $\lambda = 3$ are formalized in Lean 4 (core library only).

1 The conjecture and the clause we refute

Definition: The Nakajima monomial crystal $M(\lambda)$ denotes the highest-weight monomial graph generated by monomial variables under Chevalley rules. *Conjecture:* $M(\lambda)$ decomposes into connected components $\bigoplus B(\mu)$, where μ runs exactly over dominant integral weights bounded by λ in the dominance order, each appearing exactly once.

The Chinese version says the same (“the connected-component decomposition of $M(\lambda)$ is $\bigoplus B(\mu)$, μ ranging exactly over the dominant integral weights not exceeding λ in dominance order, each exactly once”). No Lie algebra is fixed, so the statement is universal over (symmetrizable) Kac–Moody algebras and dominant λ ; a single algebra and a single λ suffice for a disproof. We take $\mathfrak{g} = \mathfrak{sl}_2$.

For $\mathfrak{sl}_2$ the dominant integral weights are $\mu = m\varpi$ with $m \in \mathbb{N}$; we write simply $\mu = m$. The simple root is $\alpha = 2$ and $\mu \leq \lambda$ in dominance order means $\lambda - \mu \in \mathbb{N}\alpha$, i.e. $\mu \leq \lambda$ and $\lambda - \mu$ even. Hence the conjecture predicts

$$M(\lambda) \cong B(\lambda) \oplus B(\lambda - 2) \oplus B(\lambda - 4) \oplus \cdots, \quad \text{in particular} \quad M(2) \cong B(2) \oplus B(0).$$

Clause refuted: for $\mathfrak{g} = \mathfrak{sl}_2$ and $\lambda = 2$, $M(2) \not\cong B(2) \oplus B(0)$ (Theorem 4); in fact $M(\lambda) \not\cong \bigoplus_{\mu} B(\mu)$ for every $\lambda \geq 2$ (Theorem 5). If “bounded by λ ” is read as the plain inequality $\mu \leq \lambda$ without the parity condition ($\mu \in \{0, 1, 2\}$ for $\lambda = 2$), the claim still fails, and even more so: $M(\lambda) \cong B(\lambda)$ (Theorem 5) misses $B(\lambda - 1)$ for every $\lambda \geq 1$ and also $B(\lambda - 2)$ for $\lambda \geq 2$ (Lean: `main_fails_all`).

2 Definitions

2.1 The monomial crystal in type A_1

We use Kashiwara's version [1] of Nakajima's monomial crystal [2]; see also [3]. In rank one the constants c_{ij} of [1] play no role, and Nakajima's variables Y_{aq^k} are identified with Kashiwara's Y_m by re-indexing (Remark 6).

Let $\mathcal{M}$ be the set of Laurent monomials $M = \prod_{m \in \mathbb{Z}} Y_m^{y_m}$ ($y_m \in \mathbb{Z}$, finitely many nonzero), and $A_m = Y_m Y_{m+1}$. Put

$$\text{wt}(M) = \sum_m y_m, \quad \varphi(M) = \max_{n \in \mathbb{Z}} \sum_{k \leq n} y_k, \quad \varepsilon(M) = \max_{n \in \mathbb{Z}} \left(- \sum_{k > n} y_k \right).$$

Both maxima are ≥ 0 (take n below, resp. above, all indices) and $\varphi(M) - \varepsilon(M) = \text{wt}(M)$. The Kashiwara operators are

$$\tilde{f}M = \begin{cases} A_{n_f}^{-1}M, & \varphi(M) > 0, \\ 0, & \varphi(M) = 0, \end{cases} \quad \tilde{e}M = \begin{cases} A_{n_e}M, & \varepsilon(M) > 0, \\ 0, & \varepsilon(M) = 0, \end{cases}$$

where $n_f = \min\{n : \sum_{k \leq n} y_k = \varphi(M)\}$ and $n_e = \max\{n : -\sum_{k > n} y_k = \varepsilon(M)\}$. (When $\varphi(M) = 0$ the set defining n_f is unbounded below, so the formula gives no n_f ; the value $\tilde{f}M = 0$ is the one under which φ is the length of the $\tilde{f}$ -string, and it is the standard semi-normal convention (see Kang–Kim–Shin [3]; Kashiwara [1] shows that $\mathcal{M}$ is semi-normal). The same remark applies to $\tilde{e}$.) $\mathcal{M}$ is a crystal [1]; in particular $\tilde{f}M = M' \iff \tilde{e}M' = M$.

Definition 1. For a dominant weight λ , $M(\lambda)$ is the subcrystal of $\mathcal{M}$ generated by the highest-weight monomial Y_0^λ : the set of monomials obtained from Y_0^λ by finite sequences of $\tilde{e}$ and $\tilde{f}$ (equivalently, since $\tilde{e}$ and $\tilde{f}$ are mutually inverse, the connected component of Y_0^λ in the crystal graph). This is the standard meaning of the notation $M(\lambda)$ in the literature [1, 3], and it is what the conjecture's definition describes ("the highest-weight monomial graph generated ... under Chevalley rules"). We call it the *main reading*.

2.2 The target crystal

$B(\mu)$ ($\mu \in \mathbb{N}$) is the crystal of the irreducible $\mathfrak{sl}_2$ -module of highest weight μ : the string $b_0 \rightarrow b_1 \rightarrow \dots \rightarrow b_\mu$ with $\text{wt } b_k = \mu - 2k$, $\tilde{f}b_k = b_{k+1}$ ($k < \mu$), $\tilde{f}b_\mu = 0$, $\tilde{e}b_k = b_{k-1}$ ($k > 0$), $\tilde{e}b_0 = 0$. The predicted crystal $D(\lambda) = \bigoplus_\mu B(\mu)$, μ dominant with $\mu \leq \lambda$, has one copy of each such string. An isomorphism $M(\lambda) \cong D(\lambda)$ is a bijection ψ preserving weights and satisfying $\psi(\tilde{f}x) = \tilde{f}\psi(x)$ and $\psi(\tilde{e}x) = \tilde{e}\psi(x)$ (including $\tilde{f}x = 0 \iff \tilde{f}\psi(x) = 0$, and the same for $\tilde{e}$).

Lemma 2. Let $\psi : V \rightarrow D(\lambda)$ be an isomorphism. Then ψ restricts to a weight-preserving bijection between the highest-weight vertices ($\tilde{e}x = 0$) of V and those of $D(\lambda)$. The highest-weight vertices of $D(\lambda)$ are the b_0 of the copies of $B(\mu)$, one of each weight $\mu \in \{\lambda, \lambda - 2, \dots\}$. In particular V must contain exactly one highest-weight vertex of weight μ for each such μ , and no others.

Proof. $\tilde{e}x = 0 \iff \tilde{e}\psi(x) = 0$, and ψ preserves weights. In $B(\mu)$ only b_0 has $\tilde{e}b = 0$, and its weight is μ . $\square$

3 The counterexample $\lambda = 2$

Proposition 3. $M(2) = \{Y_0^2, Y_0Y_1^{-1}, Y_1^{-2}\}$ and $M(2) \cong B(2)$.

Proof. Direct computation from the definitions:

M	wt	φ	ε	$\tilde{f}M$	$\tilde{e}M$
Y_0^2	2	2	0	$A_0^{-1}Y_0^2 = Y_0Y_1^{-1} \ (n_f = 0)$	0
$Y_0Y_1^{-1}$	0	1	1	$A_0^{-1}Y_0Y_1^{-1} = Y_1^{-2} \ (n_f = 0)$	$A_0Y_0Y_1^{-1} = Y_0^2 \ (n_e = 0)$
Y_1^{-2}	-2	0	2	0	$A_0Y_1^{-2} = Y_0Y_1^{-1} \ (n_e = 0)$

For instance, for $M = Y_0Y_1^{-1}$ the prefix sums $\sum_{k \leq n} y_k$ are 0 ($n < 0$), 1 ($n = 0$), 0 ($n \geq 1$), so $\varphi = 1$, $n_f = 0$; the values $-\sum_{k > n} y_k$ are 0 ($n < 0$), 1 ($n = 0$), 0 ($n \geq 1$), so $\varepsilon = 1$, $n_e = 0$. The set is closed under $\tilde{e}, \tilde{f}$ and contains Y_0^2 , and every element is reached from Y_0^2 by $\tilde{f}$; so it is $M(2)$. The map $Y_0^{2-k}Y_1^{-k} \mapsto b_k$ is an isomorphism onto $B(2)$. $\square$

Theorem 4. $M(2) \not\cong B(2) \oplus B(0)$. Hence Conjecture 00000003824 is false.

Proof. $B(0) = \{b_0\}$ is a highest-weight vertex of weight 0. By Lemma 2, an isomorphism would need a monomial $x \in M(2)$ with $\tilde{e}x = 0$ and $\text{wt } x = 0$. The only element of weight 0 is $Y_0Y_1^{-1}$, and $\tilde{e}(Y_0Y_1^{-1}) = Y_0^2 \neq 0$. (Alternatively: $|M(2)| = 3 \neq 4 = |B(2) \oplus B(0)|$.) $\square$

4 Every $\lambda \geq 2$ fails

Theorem 5. For every $\lambda \geq 1$, $M(\lambda) = \{Y_0^{\lambda-k}Y_1^{-k} : 0 \leq k \leq \lambda\} \cong B(\lambda)$. Consequently, for every $\lambda \geq 2$, $M(\lambda) \not\cong \bigoplus_{\mu \leq \lambda} B(\mu)$, because $B(\lambda - 2)$ is missing.

Proof. Let $M_k = Y_0^aY_1^b$ with $a = \lambda - k$, $b = -k$. If $0 < k < \lambda$ ($a > 0 > b$), the prefix sums are 0, a , $a + b$, so $\varphi = a$ and $n_f = 0$, and $\tilde{f}M_k = A_0^{-1}M_k = Y_0^{a-1}Y_1^{b-1} = M_{k+1}$; the values $-\sum_{k > n}$ are $-(a + b)$, $-b$, 0, so $\varepsilon = -b = k$ and $n_e = 0$, and $\tilde{e}M_k = A_0M_k = M_{k-1}$. For $k = 0$: $\varphi(Y_0^\lambda) = \lambda$, $\tilde{f}Y_0^\lambda = M_1$, $\varepsilon = 0$. For $k = \lambda$: $\varphi(Y_1^{-\lambda}) = 0$, $\varepsilon = \lambda$, $\tilde{e}Y_1^{-\lambda} = A_0Y_1^{-\lambda} = M_{\lambda-1}$. So the set $\{M_0, \dots, M_\lambda\}$ is closed, it is a single $\tilde{f}$ -string from M_0 , and $M_k \mapsto b_k$ is an isomorphism with $B(\lambda)$. Its only highest-weight vertex is M_0 , of weight λ . For $\lambda \geq 2$, $\lambda - 2$ is dominant and $\leq \lambda$, and by Lemma 2 the predicted crystal requires a highest-weight vertex of weight $\lambda - 2$. $\square$

For $\lambda \in \{0, 1\}$ the only dominant $\mu \leq \lambda$ is λ itself and the claim holds. Theorem 5 is the $\mathfrak{sl}_2$ case of Kashiwara's theorem [1] that the connected component of a highest-weight monomial of dominant weight λ is isomorphic to $B(\lambda)$, in every type. So the conjecture fails whenever some dominant $\mu < \lambda$ exists; for example, for $\mathfrak{sl}_3$ and the adjoint weight $\theta = \varpi_1 + \varpi_2$, $M(\theta)$ has 8 elements (it is $B(\theta)$), while $B(\theta) \oplus B(0)$ has 9 (checked by `verify.py`).

Remark 6. Conventions. (i) Nakajima [2] uses variables Y_{i,aq^k} and A_{i,aq^k} with a parity condition; after re-indexing these become Kashiwara's Y_m, A_m for a suitable choice of c (in rank one there is no choice), as explained in [1]. (ii) The mirrored convention (sums over $k \geq n$ instead of $k \leq n$) is obtained by $m \mapsto -m$ and gives an isomorphic crystal. (iii) Starting from Y_s^λ instead of Y_0^λ is a shift of indices. In all cases $M(\lambda)$ is a single string $B(\lambda)$; `verify.py` checks (i)–(ii) for $\lambda \leq 8$.

5 Other readings of “ $M(\lambda)$ ”

The definition in the conjecture is informal, so we also consider the two other readings we could think of in which $M(\lambda)$ is built from monomials and has several components.

(U) Union reading. $M_U(\lambda)$ is the union of the subcrystals generated by *all* highest-weight monomials of weight λ (that is, $\varepsilon(M) = 0$, $\text{wt } M = \lambda$). This fails for *every* $\lambda \in \mathbb{N}$: Y_0^λ and Y_1^λ (for $\lambda \geq 1$), resp. 1 and $Y_1^{-1}Y_2$ (for $\lambda = 0$), are two distinct highest-weight vertices of weight λ , so $B(\lambda)$ would occur at least twice, contradicting “exactly once” (Lemma 2). (By Kashiwara’s theorem every component of $M_U(\lambda)$ is a copy of $B(\lambda)$, so for $\lambda \geq 2$ also no $B(\mu)$ with $\mu < \lambda$ occurs; `verify.py` confirms this on all highest-weight monomials of weights $0, \dots, 3$ supported on $Y_0, \dots, Y_4$ with exponents in $[-2, 3]$, finding 190, 286, 381 and 467 of them.) Any larger reading, such as all monomials of weight $\leq \lambda$, contains $M_U(\lambda)$ and fails for the same reason.

(KR) Kirillov–Reshetikhin reading. One might take for $M(\lambda)$ the monomial crystal attached to the Kirillov–Reshetikhin module with highest weight λ . In type $A_1^{(1)}$ the KR modules are irreducible as $U_q(\mathfrak{sl}_2)$ -modules, and their highest-weight monomial (for $\lambda = 2$, Y_0Y_1 in the present indexing, i.e. $Y_aY_{aq^2}$) generates $\{Y_0Y_1, Y_0Y_2^{-1}, Y_1^{-1}Y_2^{-1}\} \cong B(2)$. So this reading coincides with the main one, and $B(0)$ is again missing (Lean: `kr2.component`, `kr2.fails`).

(P) Tensor-power reading. “Generated by monomial variables” could mean: $M_P(\lambda)$ is the subcrystal generated by the products $c_0c_1 \cdots c_{\lambda-1}$ with $c_j \in \{Y_{2j}, Y_{2j+1}^{-1}\}$, each factor being a copy $\{Y_{2j}, Y_{2j+1}^{-1}\} \cong B(1)$. This set has 2^λ elements and realizes $B(1)^{\otimes \lambda}$. For $\lambda = 2$ it is $B(2) \oplus B(0)$ (highest-weight vertices $Y_0Y_2, Y_1^{-1}Y_2$), so under this reading the case $\lambda = 2$ is *true*. For $\lambda = 3$ it has the three highest-weight vertices $Y_0Y_2Y_4$ (weight 3), $Y_0Y_3^{-1}Y_4$ and $Y_1^{-1}Y_2Y_4$ (both of weight 1), i.e. $M_P(3) \cong B(3) \oplus B(1)^{\oplus 2}$: $B(1)$ occurs twice, so “each exactly once” fails (Lemma 2). In general the multiplicities are the Clebsch–Gordan numbers (`verify.py`, $\lambda \leq 5$).

Corollary 7. *For $\mathfrak{sl}_2$ and $\lambda = 3$ the conjecture fails under the main reading, under (U) and under (P).*

6 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only).

- **Mon:** a monomial as a list of pairs (m, y_m) , $y_m \neq 0$, indices strictly increasing. `muly`, `mula` multiply by Y_n^d and A_n^s ; `wt`, `phi`, `eps` implement the formulas above (prefix/suffix maxima, the empty prefix/suffix included); `firstHit`, `lastHit` compute n_f, n_e ; `fT`, `eT` are $\tilde{f}, \tilde{e}$ with `none` for 0.
- **Reach s x:** x is obtained from s by finitely many $\tilde{f}, \tilde{e}$. `Mmain 1 = Reach (Y0pow 1)` is $M(\lambda)$. `closeN` computes closures, and `mem_closeN_iff` proves that a closed computed closure is exactly the generated subcrystal.
- **Target 1, wtB, fB, eB:** the crystal $\bigoplus_\mu B(\mu)$ with vertices (μ, k) , μ dominated by λ (Dominated: $\mu \leq \lambda$, $\lambda - \mu$ even), $k \leq \mu$. `CrystalIso` is a bijection preserving `wt`, $\tilde{f}$, $\tilde{e}$; `Decomposes V 1` is the existence of such an isomorphism onto **Target 1**; `Conjecture` is $\forall \lambda$, `Decomposes (Mmain 1) 1`.
- **M2.eq:** $M(2)$ is exactly the three monomials. `iso_M2_B2`: $M(2) \cong B(2)$.
- `conjecture_00000003824_clause_false` proves

$$\neg \text{Decomposes (Mmain 2) 2},$$

and `conjecture_00000003824_false` proves $\neg \text{Conjecture}$.

- `main_fails_all`: $\neg \text{Decomposes } (\text{Mmain } 1) \ 1$ for every $\lambda \geq 2$, via the symbolic computation of $\tilde{f}, \tilde{e}$ on the string `(fT_s1, ..., InStr_hw)`.
- `main_fails_all'`: the variant `Target'` without the parity condition fails for every $\lambda \geq 1$; `kr2_fails`: the KR reading at $\lambda = 2$.
- `union_fails` (every λ), `prod3_fails` and `all_readings_fail_at_3` for the readings (U), (P).
- Non-vacuity: `main_holds_0`, `main_holds_1`, `prod_holds_2` construct explicit isomorphisms, so `Decomposes` is satisfiable and agrees with the cases where the claim is true. `sanity` checks canonical form, $\varphi - \varepsilon = \text{wt}$ and $\tilde{e}\tilde{f} = \text{id}$, $\tilde{f}\tilde{e} = \text{id}$ on all components used.

There is no `sorry`, `native_decide` or added axiom. The command `#print axioms` shows only the standard axioms `propext` and `Quot.sound`; in addition, `Classical.choice` appears for the three theorems `main_fails_all`, `main_fails_all'` and `all_readings_fail_at_3`.

Scope. The Lean development is for $\mathfrak{sl}_2$. It does not formalize Kashiwara’s general theorem; it computes the relevant components directly. The crystal axiom $\tilde{f}M = M' \iff \tilde{e}M' = M$ is checked on the components used but not proved for all monomials; it is only needed to identify the subcrystal generated by Y_0^λ with the undirected connected component, which is a standard fact [1]. The $\mathfrak{sl}_3$ example and the enumerations quoted for (U) and (P) are in `verify.py` only.

7 Independent check

`verify.py` (Python 3 standard library) implements the monomial crystal for an arbitrary Cartan matrix with Kashiwara’s constants c_{ij} . It computes φ , ε , n_f , n_e by brute force over all n in a window around the support, which is a different method from the scans in Lean. It checks: $M(\lambda)$ is a single string $B(\lambda)$ for $\lambda \leq 8$ in both conventions of Remark 6(ii), and the conjecture holds exactly for $\lambda \leq 1$; the union reading (U); the tensor-power reading (P) with Clebsch–Gordan multiplicities for $\lambda \leq 5$; and $|M(\varpi_1 + \varpi_2)| = 8$ for $\mathfrak{sl}_3$.

8 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # standard library only
pdflatex report.tex
```

References

- [1] M. Kashiwara, Realizations of crystals, in: *Combinatorial and Geometric Representation Theory*, Contemp. Math. 325, AMS, 2003, 133–139.
- [2] H. Nakajima, t -analogs of q -characters of quantum affine algebras of type A_n , D_n , in: *Combinatorial and Geometric Representation Theory*, Contemp. Math. 325, AMS, 2003, 141–160.
- [3] S.-J. Kang, J.-A. Kim, D.-U. Shin, Monomial realization of crystal bases for special linear Lie algebras, *J. Algebra* 274 (2004), 629–642.